

MOHAMMAD MILHEM

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EDUCATION

Jordan University of Science and Technology

Bachelor of Science in Aeronautical Engineering

Faculty Honor List, 4 consecutive semesters

Relevant Coursework: Fluid Mechanics, Thermodynamics, Statics, Dynamics, Mechanics of Materials, Solid Modeling

Irbid, Jordan

Expected June 2029

GPA: 4.0/4.0

PROJECTS AND LEADERSHIP

JUST Formula Student Team (BARQ Racing) — Team Lead & Head of Aerodynamics

August 2026 – Present

- Lead a 15-member team developing a formula-style race car for the Formula Student UK Concept Class.
- Built the team's operating infrastructure from the ground up, including documentation, workflows, and standards that keep all sub-teams aligned on schedule and quality.
- Direct a 3-member aerodynamics department developing the front wing, rear wing, undertray, and diffuser.

Aerodynamic Force Measurement Rig — Independent Project

July 2026 – Present

- Designed a lift and drag balance using load cells and an MPXV7002DP differential pressure sensor for airspeed, with Arduino Uno data acquisition and a live display.
- Developing a validation process comparing measured forces against ANSYS Fluent simulations of the same CAD models, to guide a second-version design.

CFD & FEA Analysis of a NACA 2412 Wing — Independent Project

August 2026

- Coupled ANSYS Fluent and ANSYS Mechanical in Workbench to study how aerodynamic lift deflects an aluminum 6061-T6 wing.
- Built a half-span symmetric CFD domain with local mesh refinement and ran an SST k- ω simulation at 30 m/s, predicting 88.1 N of lift.
- Mapped the CFD surface pressure onto the structural model as a distributed load, giving a maximum tip deflection of 0.117 mm.

Boeing 747-8 Full-Aircraft CAD Model — AE204 Solid Modeling

April – May 2026

- Modeled and assembled a complete 747-8 in Creo Parametric, including fuselage, wings, raked wingtips, empennage, GENx engines, and landing gear, in three weeks.
- Built the fuselage from scaled blueprint cross-sections using boundary blends, and the wing from three airfoil sections where public data was unavailable.
- Self-taught techniques beyond the course, including surface projection and offset features, to add windows, doors, and a patterned 18-blade fan.

C-130 Hercules Analysis & Standard Atmosphere Model — AE201 Team Project

December 2025 – January 2026

- Co-authored a 41-page technical report on the C-130's design, structures, propulsion, performance, and cost; led the history, evolution, and modernization sections.
- Co-developed a MATLAB model of the 1976 U.S. Standard Atmosphere up to 105 km, computing temperature, pressure, density, and speed of sound, with plots and Excel export.
- Stored all layer constants in a single matrix, with base temperatures and pressures computed in a loop rather than hard-coded; validated outputs against standard tables at 20 km (within 1%).

HONORS AND AWARDS

Faculty Honor List, Jordan University of Science and Technology

Queen Rania Outstanding UK Learner Award

Fall 2024 – Spring 2026

2024

CERTIFICATIONS

ANSYS Workbench: Fluent & Mechanical (45 hours) — MechXpert

August 2026

SKILLS

Software: ANSYS Fluent, ANSYS Mechanical, Creo Parametric, MATLAB, Microsoft Excel

Analysis: CFD, FEA, meshing, coupled fluid-structure analysis, surface and assembly modeling

Hardware: Arduino, load cells, differential pressure sensors, data acquisition